MSc ARTIFICIAL INTELLIGENCE • DISSERTATION SUMMARY

The Emergence of Internal Representations in Sequence Models

Whether a JEPA objective helps depends on the architecture: Transformer vs Mamba on Othello

Building a JEPA objective for discrete action sequences, and testing it against next-token training

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The question

Two things decide what a sequence model learns: the architecture that carries the history forward, and the objective that supplies the learning signal. Most work varies one and holds the other fixed. Both are varied here, across three environment scales, and one objective had to be built before it could be tested. JEPA comes from vision, and where it has reached discrete tokens (LLM-JEPA) it sits on top of next-token training. As the sole objective in action sequences it had no working recipe. The testbed is Othello, where the true state is known exactly at every step, so architecture and objective can be compared not only on whether a model predicts well but on whether it has internally represented an environment it was never shown.

Building a JEPA objective for discrete actions

The failure mode took five versions to locate. With **nested views**, where the context and target encoders share the game history, the predictor can lean on the shared prefix instead of learning anything about the future. Nothing fixed it from inside that design: an EMA target encoder, longer horizons, weaker and stronger predictors, multi-position prediction and InfoNCE in place of regression all left top-1 legality between 25% and 40%, and one variant collapsed.

Two changes fixed it, and the numbers show that neither was sufficient alone:

Target view	Supervision	Top-1 legal
Nested (shares the prefix)	every valid position	25.4%
Hard-disjoint (future only)	one sampled position	73.3%
Hard-disjoint (future only)	every valid position	**99.0%**

Removing the prefix from the target forces the predictor to work from what the context encoder actually extracted; supervising every valid prefix boundary rather than one sampled point densifies the signal. The EMA target encoder, which changed nothing inside the nested design, then took it to 99.4%. (These are development runs on the 8×8 Transformer under the earlier evaluation protocol, recording how the design was reached rather than the final numbers below.) The final recipe is a **hard-disjoint target, an EMA target encoder, InfoNCE at temperature 0.1, and a linear** $512\to512$ **predictor**, with the loss computed at every valid position. On Mamba the same configuration reached 99.7% legality and 98.2% linear decoding of the board state.

Design

	Autoregressive	JEPA-like
Transformer	baseline	new
Mamba (SSM)	alt. architecture	fully new

Each cell trained at 8×8, 12×12 and 16×16. Parameters matched to within 0.86% (25.3M vs 25.5M), shared data and hyperparameters, ∼20M games per condition, 54 GPU-hours, five seeds at the extreme scales.

Each of the twelve systems is measured on three things the literature often collapses into one:

- **Behaviour**: is the predicted move legal? (top-1, P@k, probability mass on the legal set)
- **Representation**: can the board state be read out of the model? (linear and MLP probes at every layer)
- **Causal use**: does editing the represented state change the prediction as the rules require? (three intervention protocols against a geometry-matched random control)

What came out

	Top-1 legal (%)		Decoding (%)	
	8×8	16×16	8×8	16×16
Transformer-AR	99.73	98.99	96.91	82.88
Transformer-JEPA	99.42	97.54	96.72	**78.37**
Mamba-AR	**99.85**	**99.05**	97.85	83.35
Mamba-JEPA	99.72	98.41	**98.04**	**83.77**
spread	*0.43*	*1.51*	*1.32*	*5.39*

All values are the main run (seed 42). Decoding is macro-balanced accuracy of a linear probe on the relative board state. Bootstrap CIs span under 0.03 pp, and the ranking is reproduced across five independent seeds. Full twelve-condition tables, per-layer probe curves and the statistical tests are in the dissertation.

Causal use. The adapted intervention edits the represented board and measures how much probability moves onto the edited board's legal moves. It raised that mass in all 12 conditions: by 5.97–6.37 pp at 8×8, 2.19–2.72 pp at 12×12 and 0.78–1.48 pp at 16×16, and it exceeded a geometry-matched random control in all 12. The effect shrinks with board size. Only this method had its strength tuned on a validation split.

1. The objective's effect depends on the architecture. The same JEPA-like objective *improves* Mamba's internal representation and *degrades* the Transformer's. Mamba-JEPA is the best of all four systems at decoding, Transformer-JEPA the worst. Averaged over architectures the objective looks mildly harmful, which is an artefact of averaging two opposite responses. Estimated across five independent training runs rather than the single run tabulated above, the matched interaction term reaches +5.09 pp at 16×16 (95% CI +4.23 to +5.95); on the single seed in the table it is +4.93. A study varying one factor would have reported the wrong main effect.

2. Predicting in representation space nearly matches predicting the token. The JEPA systems never train on next-move prediction and read out through a head fitted afterwards, yet land within 0.3 pp of autoregressive training at 8×8 and within 0.6 on Mamba at 16×16. Autoregressive training remains the stronger choice for legality, as expected of an objective that optimises it directly, though the gap is narrower than that framing suggests.

3. The standard benchmark conceals these differences. The spread on decoding between best and worst system grows from 1.32 to 5.39 pp as the board scales, while top-1 legality stays above 97.5% throughout. At the 8×8 scale the whole field works at, the interaction above is essentially invisible.

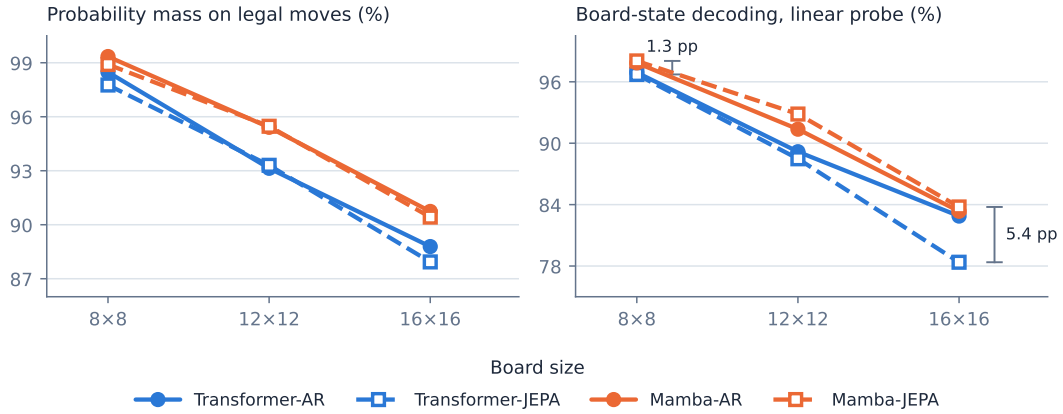

Figure 1: Colour encodes architecture, line style encodes objective. The two Transformer variants separate at 16×16 while the two Mamba variants stay together: the interaction is visible only off the saturated scale.

What this does not show

The JEPA development was a search for a working configuration, not a controlled ablation: some early versions changed predictor, training budget and evaluation protocol at once, so it documents how the design was reached rather than isolating each factor. The architecture comparison is confounded with depth, because matching parameters forced 8 Transformer blocks against 15 Mamba blocks, which no parameter-matched design can avoid. The two objectives do not use identical prediction heads, so the behavioural comparison is between complete systems rather than encoders; the probe results carry no head and are free of this. Training data are uniformly random legal games, so the models learn the task's constraints, not strategy.

Where this goes next

The obvious next step is the ablation the dissertation did not run: hold the budget fixed and vary one element of the JEPA configuration at a time (view construction, number of supervised positions, predictor capacity, loss) so that the recipe becomes a set of design rules rather than one working point. Several branches explored and set aside belong here too: conditioning the predictor on the action, treating order-permutations that reach the same state as positives, and multi-step rollout in representation space.

The second direction is where the two axes meet. An objective is not good or bad on its own, only in combination with a mechanism for carrying history, so the practical question is when a linear-complexity architecture starts paying off, at which sequence length and parameter budget, and whether the objective that suits it differs from the one that suits attention. The measurements here give one point on that curve and a protocol for getting more.

Third: the same evaluation triad should transfer to environments with partial observability and stochastic transitions, where what a model ought to represent is far less clear.